

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 50

Question Count: 5

Code of Conduct

I _____ Kanmani L(192511414)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____



Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

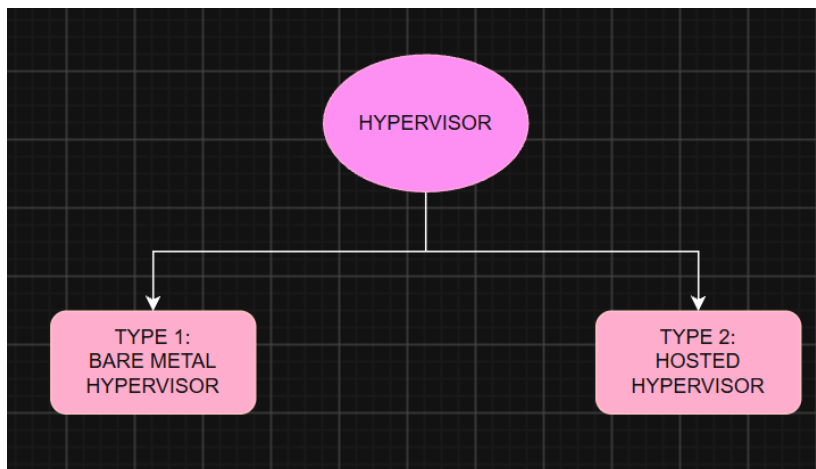
Answer:

Hypervisor:

A hypervisor, also called a Virtual Machine Monitor (VMM), is software that creates and manages virtual machines (VMs) on a physical server. It allows multiple operating systems and applications to run on the same physical hardware.

There are two major types of VMMs:

1. Bare metal hypervisor, also known as type-1 hypervisor
2. Hosted Hypervisor , lso known as type-2 hypervisor



Comparison between type 1 and type 2 hypervisor:

Aspect	Type 1	Type 2
Installation	Easier To Install As It Runs Directly On The OS	It Runs On Top Of The Os
Performance	High	Low Due To OS Overhead
Resource Management	Direct Hardware Access	Accesses Hardware Through Host OS
Reliability	High	Comparatively Lower

Performance Analysis:

- Type-1 hypervisors directly interact with the hardware. Therefore, CPU, memory, storage, and network resources can be allocated efficiently to VMs. This results in low overhead and high performance.
- Type-2 hypervisors depend on a host operating system. Requests from a VM pass through the hypervisor and host OS before reaching hardware. This introduces additional overhead and may reduce performance.

Security Analysis:

- Type 1 has more security as compared to the type 2, as type 1 directly runs on the OS, they are not involved in any other hardware
- But in the case of type 2, the hypervisor is placed on the top of the OS, thus, it undergoes interfering with multiple other software

Manageability:

- Type 1 is relatively easier to manage compared to the Type 2 Hypervisor
- Bare metal hypervisor acts as an OS, without the need of another software. But hosted rely on other OS, thus management of other OS is required

Justification:

For a cloud datacenter hosting mission-critical applications, Type 1 is more suitable, as they are:

Reliable

Manageable

Shows Better Performance

Has More Security Aspects As Compared To The Type 2

2.A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Answer:

A cloud provider may run workloads belonging to different organizations on the same physical server. For example, finance, healthcare etc and may share the same physical infrastructure. Since these workloads contain sensitive information, strong protection of data is essential.

- Virtualization creates isolated virtual environments called Virtual Machines (VMs) on the same physical hardware.

How isolation in virtualization works:

Each workload can run in a separate VM.

For example:

- VM 1 for Finance applications
- VM 2 for Healthcare applications
- VM 3 for Student portal

The hypervisor separates the VMs and allocates dedicated CPU, memory, storage, and network resources.

Memory Isolation:

- The hypervisor maintains separate memory spaces for each VM. One VM cannot directly access the memory allocated to another VM.
- This prevents sensitive healthcare or financial information from being accessed by other workloads.

CPU Isolation

CPU resources can be allocated using resource limits and reservations. This prevents one workload from consuming all CPU resources and affecting other applications.

Storage Isolation

Separate virtual disks and storage volumes can be created for each workload. Access controls and encryption can protect sensitive data.

Network Isolation

Virtual networks, VLANs, VXLANs, and software-defined networking can separate network traffic between different workloads.

Attacks in isolation:

IN HYPERVISOR:

If an attacker exploits a vulnerability in the hypervisor, they may attempt a VM escape attack, where they move from one VM to another or access the host system.

How to prevent:

Secure configuration

Access control

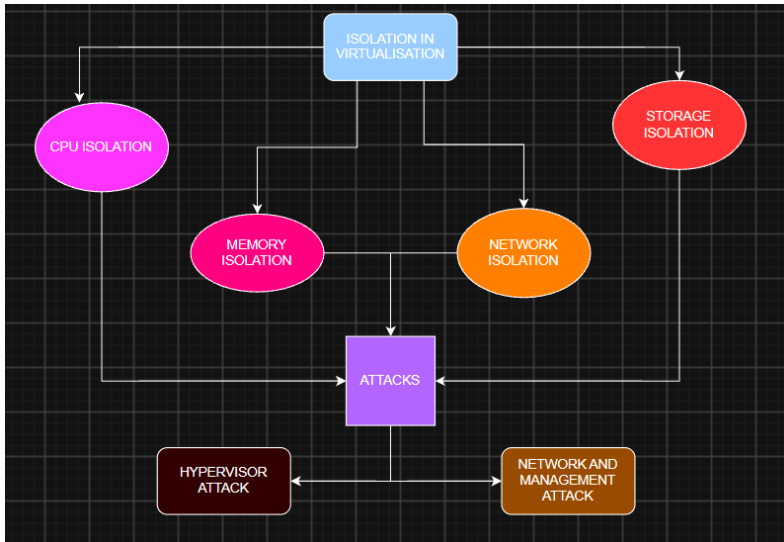
Monitoring

IN NETWORK AND MANAGEMENT:

An attacker who gains access to management systems may modify VM configurations or access sensitive resources.

How to prevent:

- Strong authentication
- Multi-factor authentication
- Encryption
- Role-Based Access Control (RBAC)
- Continuous monitoring



Conclusion:

Virtualization provides strong isolation through separate VMs, memory protection, storage separation, and network segmentation.

However, the hypervisor and management/network interfaces remain major attack surfaces.

A combination of secure hypervisor configuration, encryption, segmentation, access control, and continuous monitoring is required to protect finance, healthcare, and student workloads.

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

Answer:

given:

CPU:92%

Memory:88%

Storage I/O Latency :35 Ms

Average VM Boot Time:170 S.

This shows High CPU and memory utilization combined with high storage latency

Analysis of the given aspects:

1. CPU utilization: 92%

CPU utilization of 92% is very high. This indicates that the physical host is close to its processing capacity.

Possible causes include:

- Too many VMs running on the host
- CPU-intensive applications
- Background processes

When CPU resources are insufficient, VMs experience CPU process disruption and increased response time.

Memory utilization – 88%

Memory utilization of 88% indicates that most physical RAM is already occupied.

This can cause:

- Memory contention
- Swapping
- Reduced application performance

If the host starts swapping memory to disk, performance may decrease significantly.

Storage I/O latency– 35 ms

A storage latency of 35 ms is relatively high for many VM workloads. This means that VM read/write requests take longer to complete.

Possible causes:

- Overloaded storage devices
- Shared storage contention

- Slow HDD-based storage

High storage latency can directly contribute to slow VM boot times.

VM Boot Time – 170 seconds

An average boot time of 170 seconds is high. The most likely causes are resource contention and slow storage I/O.

During boot, a VM requires CPU, memory, and storage operations. Since all three resources are under pressure, VM startup becomes slow.

Corrective Measures

Increase CPU Capacity

- Migrate some VMs to another host.
- Add more physical CPU resources.

Increase Memory

- Add more RAM to the physical host.
- Optimize VM memory allocation.
- Reduce memory overcommitment.

Improve Storage

- Move VMs to SSD storage
- Use storage caching where appropriate.
- Distribute I/O-intensive workloads across storage systems.

Optimize VM Boot Process

- Reduce unnecessary startup services.
- Use VM templates.
- Use faster storage.

Monitor and Balance Workloads

Continuous monitoring should be implemented to detect resource utilization.

Conclusion

The high CPU utilization, high memory utilization, and storage I/O latency constraints are the primary reasons for the high VM boot time of 170 seconds.

The best solution is to rebalance workloads, increase CPU and memory capacity, upgrade storage, and implement continuous monitoring and automated VM migration.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations

Answer:

A Software Defined Datacenter uses virtualization and software-based management to control datacenter resources.

- It virtualizes compute, storage, and networking, allowing infrastructure to be maintained and managed using software.
- Traditional datacenters depend heavily on physical hardware configuration, while SDDC provides centralized control.

Difference Between Traditional Datacentre And Sddc:

Aspects	Traditional Datacenter	Sddc
Storage	Physical Storage	Software Storage
Provisional	Manual	Automated
Scalability	Slow	Can Expand And Rapid
Maintanence	Low	High
Reliable	Less	More

Computer virtualisation:

In SDDC, physical servers are Installed using hypervisors. Multiple VMs can run on a single physical server.

Benefits:

- Rapid VM provisioning
- Better hardware utilization
- High availability

if an application requires additional resources, CPU and memory can be dynamically allocated without physically installing a new server.

Storage virtualization:

Storage virtualization combines physical storage resources into a flexible virtual storage.

Benefits:

- Centralized storage management
- Dynamic storage allocation
- Easier scaling
- Improved availability

Network virtualisation:

Network virtualization separates network services from physical networking hardware.

Technologies such as VLAN allow administrators to create and modify virtual networks through software.

Benefits:

- Faster network configuration
- Improved security
- Easy workload mobility

How SDDC Improves Agility

Automated Provisioning

- New VMs, storage, and networks can be created quickly using software and tools.

Faster Scaling

- Resources can be increased or decreased according to demand.

Centralized Management

- Administrators can manage the entire infrastructure from a centralized management platform.

Reduced Hardware Dependency

- Infrastructure can be controlled through software instead of manual work

Improved Disaster Recovery

- Virtual machines and data can be replicated and moved between physical systems more easily.

Therefore, SDDC improves datacenter agility by virtualizing compute, storage, and networking. It enables automated provisioning, rapid scaling, centralized management, and efficient resource utilization.

Compared with traditional datacenters, SDDC provides a more flexible, scalable, automated, and responsive infrastructure.

5.A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Answer:

A multi-cloud environment uses services from multiple cloud providers, such as AWS, Microsoft Azure, and Google Cloud.

Cloud federation allows users and applications to access resources across different cloud providers using trusted identity relationships.

In this case, the federation project fails because different cloud providers must have used different identity formats, authentication mechanisms, and authorization models.

Analysis:

Identity mismatch may occur because of differences in:

- User IDs
- Role definitions
- Authentication protocols
- Attribute names
- Access control

For example, a user may be identified by an email address in one cloud and by a unique user ID in another cloud.

Similarly, a role such as "Admin" in one provider may have completely different permissions from the same role in another provider.

This prevents users from being correctly authenticated and authorized across multiple clouds.

Thus it is very important to maintain a secure identity design in all platform which will further reduce the risk of identity mismatch and prevent project failure which may cause greater financial loss.

How to create Secure Identity Design

Central Identity Provider

A centralized Identity Provider should manage user identities.

The organization can use a trusted identity system, which is common and does not mismatch with other resources identity

The cloud providers act as Service Providers that trust the central identity provider.

Authentication:

Use standard federation protocols

These protocols allow users to authenticate once and securely access authorized services across multiple clouds.

By following the protocols, the authentication failure will be neglected.

Single Sign-On

Single Sign-On (SSO) allows users to authenticate through the central identity provider instead of maintaining separate credentials for every cloud.

This reduces password-related risks and improves user experience.

Identity Mapping

A common identity schema should be created to map users and roles between providers.

This ensures that permissions are correctly translated across different cloud environments.

Role-Based Access Control

Use RBAC to assign permissions based on job roles.

- Student -Access student portal
- Faculty - Access academic systems
- Administrator- Manage cloud resources

Encryption

Identity information and communication should be protected using encryption.

All authentication and authorization activities should be logged and monitored to detect suspicious behaviour.

The central identity provider authenticates the user. Federation protocols securely transfer authentication information to the required cloud provider. The provider then maps the user's identity and role to appropriate permissions.

Conclusion:

The multi-cloud federation failure is mainly caused by incompatible identity formats, authentication mechanisms, and authorization models. A secure solution is to implement a central Identity Provider, standardized identity mapping, RBAC, least privilege, encryption, etc. This provides secure identity management across multiple cloud providers while protecting user privacy.